Innovative Assignment 1

SMART CPU SCHEDULER

Objective

Design and implement a CPU scheduling simulator that can schedule processes and compare different scheduling approaches. The goal is to apply CPU scheduling concepts to a practical problem instead of only solving Gantt-chart questions manually.

What You Have to Do

• Create a program that accepts process details such as Process ID, Arrival Time, Burst Time and Priority.

• Implement at least **three standard algorithms**: FCFS, SJF/SRTF, Priority and/or Round Robin (choose any three or more).

• Your program must generate the **Gantt Chart** automatically.

• Calculate and display Average Waiting Time, Average Turnaround Time and Average Response Time. You may also show CPU Utilization and Context Switches.

• Test the algorithms using at least **3 different workloads** (for example: short jobs, mixed jobs and high-priority jobs).

• Write a short comparison explaining how the algorithms behave for the selected workloads.

• Add one meaningful additional feature, such as dynamic priority, aging, workload-based algorithm selection, random process generation, GUI/visualization, or another justified idea.

What You Must Submit

• Source code with clear comments.

• A one-to-two page report containing algorithm explanation, test cases, screenshots/results and comparison.

• At least three test cases with output.

• Gantt charts and calculated performance measures.

• A short explanation of the additional feature implemented.

Important: The assignment must demonstrate your own understanding. Do not submit copied code or copied project reports.